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DISCURSIVE ARTICLE

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GrID (GastroIntestinal Disease): An interactive Sudoku-based clinical anatomy card game for first-year medical students

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Abstract

Application of anatomical knowledge is an essential component of both clinical practice and medical education. Within dissection room–based practicals, game-based learning was introduced to a first-year undergraduate medicine session and designed to creatively engage students, cement understanding of the embryological subdivisions of the gastrointestinal tract, and encourage association with clinical acumen. A novel interactive card game established from the number game Sudoku was constructed to incorporate both practical-specific and wider course learning objectives. 81 laminated cards were created to depict anatomical features, gastrointestinal symptoms, abdominal examination components, and sites of referred pain. Students were given a 9 ×9 square base to organize these cards into rows, columns, and boxes so that each contained three foregut, midgut, and hindgut features. An internal challenge was inserted within the game to inspire participation, which bolded the first letter of several cards to spell out the word "digestive". Teams submitted their entries as a group, and a winner was randomly selected from the correct answers. The winning teams earned a low-stakes but appropriate prize—a packet of chocolate digestive biscuits. Highlights of this resource include the reproducibility and sustainability of the format, which required minimal setup and equipment. On a local level, valuable insights were gained into student social dynamics and their effects on engagement. Observed competitive behaviors inspired reflections around professionalism and motivational factors within a largely formative learning landscape. Recommendations for future implementation of this resource explore its use among other anatomical systems and delivery across varied cohorts.

KEYWORDS
clinical anatomy, game-based learning, medical education, Sudoku

INTRODUCTION

The plethora of anatomical teaching methods available continues to expand with advancing technology and educational research. While delivery may be dependent on resources and allocated time available, a combination of approaches remains the most effective.[1] Within this lies the expanding field of game-based learning and the benefits and challenges it brings to anatomy education.

Game-based learning has been described as the process of including bespoke game elements alongside other teaching formats in order to increase student engagement and learning.[2] This is distinct from the similarly titled gamification, which adds gaming processes to existing teaching, such as awarding points and prizes to multiple-choice questioning.[3] Notably, these terms are often used interchangeably within anatomy education literature, and indeed within individual studies.[4] Therefore, drawing understanding of the

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differing roles and educational value of these methods in this field is complex and would benefit from further exploration. Within wider educational contexts, game-based learning has been credited as a means of increasing student enjoyment of learning and developing critical thinking skills.[5] Practically speaking, the development of games is time and resource intensive, yet the sharing of experiences and tools can help inspire instructor confidence and creativity.

This discursive article will review a novel form of game-based learning within clinical anatomy education from the perspective of underlying educational theory, practical approaches, and recommendations for future application.

Anatomy games pedagogy

Anatomy teaching within medical education is primarily focused on the initial (typically preclinical) years of undergraduate studies. As a discipline, anatomy has traditionally been associated with high stress levels due to the large volume of unfamiliar learning material to be covered within this time period and nationally decreasing hours of teaching contact.[6] Integration of clinical relevance from an early onset is an important method of enhancing the meaning students assign to their learning and developing their application skills.[7]

There have been several examples of game-based learning within anatomy education to date, with noted positive student and staff feedback.[8–10] Formats have varied from high-technology simulations to digital scenario-based games and classic board game styles. Many of these take a multiplayer or team-based approach to gaming, yet utilize the principles of individual student-centered learning by creating an environment for students to become active drivers of their learning experience and enabling educators to adopt a facilitatory role.[11] This model offers a distinct benefit to large-scale anatomy teaching that often requires independent learning as part of structured practical sessions. Anatomy games that prioritized student-centered learning within their design have described increased levels of student satisfaction due to the collaborative nature of the task.[12]

To further appreciate the observed benefits of student socialization and collaboration in anatomy and inform the implementation of game-based learning, several educational theories were considered. By engaging in team games, students develop a collective goal to achieve the required objectives. This can be described as the formation of a community of practice. First defined by Lave and Wegner,[13] members of a shared community contribute their individual skillsets to the mastering of a common domain of interest, in this case, the successful completion of an educational game. Although the game may end once delivered, these students will continue to work together throughout their anatomy and medical education and, to some extent, as clinical practitioners. Therefore, the relationships developed from creating these communities of practice may extend beyond the learning activity or inspire future collaborative efforts.

Another relevant learning theory that can be seen in game-based learning is Vygotsky's zone of proximal development. This describes the subsect of skills that a learner can only demonstrate with guidance, as they aim to master independent competence.[14] This guidance can come from educators but may also be given by peers or the resources within an environment, such as games. Application of anatomical theory into clinical practice can prove challenging as it requires knowledge transformation,[7] and both games and the interactions gained from playing them can provide a different perspective that may allow students to move into the zone of proximal development.

This pedagogical perspective can be supported by evidence that both team-based learning and collaborative forms of assessment have been shown to increase anatomical knowledge.[15,16] Students who engaged in cooperative learning significantly improved their individual scores on anatomy exams, highlighting that opportunities for cooperative learning can beget academic advantage. Limited studies that included analysis of anatomy test scores before and after game-based learning have also noted improved student performance.[12,17] However, Lorenzo-Alvarez et al.[18] found that their initial positive results were not reflected in a subsequent assessment three months later, suggesting that the game elements may not benefit long term retention of anatomical knowledge. A potential downside of a cooperative approach is the risk of "social loafing" by weaker students who allow colleagues to take control of group-based activities. Green et al.[19] noted that anatomy collaborative assessments can improve team scores, but also result in a decrease in individual success due to this phenomenon. This is also an important consideration in game-based learning, as students work together with different levels of knowledge, motivation and interest in game activities. Research around the use of game-based learning in anatomy centers heavily around student perceptions, rather than performance and future studies into the potential knowledge benefits would help guide implementation.

Sudoku

The umbrella term of game-based learning includes multiple formats, such as digital, immersive, and outlay games. Outlay games are a form of card game that challenge students to identify and order components based on a larger picture.[20] While they may seem traditional when viewed alongside more advanced techniques, card-based games have been suggested to better target direct interpersonal communication among students, as they interact with each other and the cards.[21] The three-dimensional structure of outlay games, and the opportunity to categorize and classify information given, were identified as desirable features of an interactive anatomy game and have been shown to be well received by students when used in similar formats.[17] Yet to introduce an additional element of methodical thinking and application of knowledge to a well-known format, the structure and rules of the game Sudoku were adapted.

Sudoku is a number-based game that focuses on the organization of thinking within a grid system. First debuted in an edition of Dell Pencil Puzzles and Word Games in 1979,[22] it is recognized internationally as a test of logical thinking. A traditional Sudoku puzzle consists of 81 cells formed by nine rows and nine columns. Boxes are made containing nine cells each, and the objective is to ensure each box/row/column has only one copy of the digits 1–9. Playing of sudoku has shown to activate the prefrontal cortex, an area responsible for higher-order thinking and decision-making.[23] Games built around existing structures are more common[24] and are easier to understand by students who bring their own experiences into gameplay.[4] Due to the familiar nature of Sudoku by the masses and the targeting of skills that are vital in anatomical and clinical practice, it was decided to adapt the traditional Sudoku game to fit within our teaching.

This is not the first use of adapted Sudoku within education. In addition to its classical use in statistics teaching,[25,26] it has also been modified to replace the numbers with symbols or words in fields such as computer science,[27] economics[28] and chemistry.[29] Even in medicine, there has been use of lettered Sudoku puzzles to sort medical terminology.[30] However, this is a technique not previously used in anatomy education, nor expanded beyond individual letters. This may be due to the more restrictive nature of the game and difficulty in aligning anatomical features with the parameters created by an 81-cell grid. However, the three classifications inherent to the teaching of gastrointestinal embryology presented a unique opportunity to utilize this game design with minor adaptations. Furthermore, as a large component of clinical anatomy education involves the creation of connections between theory and practice, the potential for skill development in pattern-recognition and application of systematic logic afforded by Sudoku also inspired its use over alternative formats.

Goals and objectives

This game was designed to take place as part of a practical anatomy session for Year 1 undergraduate medical students at University of Bristol. As part of the overarching module structure, students spent 2 weeks reviewing a selected bodily system through case-based learning. Each system contains a dissection room-based anatomy session, which incorporates prosected cadaveric materials, anatomical models, and radiological images. In order to facilitate rotation of all 273 students through our dissection room, each session was run identically three times, containing approximately 90 students per practical. Students were then further subdivided into small groups of 10–12 and rotated in these groups through four stations. This layout lent itself well to inclusion of a game-based learning task, as use of larger groups can make it difficult for all students to participate.[17]

A challenge identified in the running of these practicals surrounded the fourth station, which combined radiology and clinically applied teaching. Within the time allotted (30 min per rotation), students would split their time between these two learning activities, meaning clinical teaching needed to be succinct and clearly emphasize the required anatomical knowledge. We soon also learned that the radiology component was inherently more didactic for a cohort relatively new to identification of anatomy on imaging. This made it logistically difficult for the single demonstrator allocated to this station to concurrently provide any meaningful teaching of clinical anatomy. Game-based learning that requires the instructor to be involved while teaching has also previously received negative staff feedback due to the lack of sole focus on either gaming or teaching activities.[6]

As a result of these reflections, a niche was identified for an independently run student activity. The session "Applied Gastrointestinal (GI) Anatomy" was chosen for a piloting of a novel clinical anatomy game, as it was the least adapted from the traditional practical layout as described above. The learning objectives of the session and the bespoke Sudoku game developed are summarized in Figure 1.

METHODS

Game design

A key principle of Sudoku is the sorting of components into boxes/rows/columns of nine, without repetition. To reduce the amount of varied components present and concentrate student learning, this was further subdivided to mirror the three embryological components of the gastrointestinal tract: the foregut, the midgut, and the hindgut. This aligned with recommendations that advocated for the use of a focused theme as opposed to generalized trivia within educational games.[31]

Each box/row/column in our adapted version now had to contain three features related to each component, creating the desired total of nine features. This is demonstrated visually in Figure 2.

Based on the learning outcomes for the session, emphasis was placed on anatomical knowledge, such as viscera, specific structural features, blood supply, and innervation. Clinical components included within this were regions of referred pain and specific symptoms of gastrointestinal disease. Some structures were included more than once, in order to emphasize the presence of features across multiple components. The name GrID (GastroIntestinal Disease) was chosen to echo the grid structure of the game. Given the time constraints for students to complete this activity, a smaller challenge was embedded within the cards created. The initial letter found on ten cards was bolded to spell out a code word. In keeping with the theme of the session, this was "digestive". An example completed version of the GrID game designed is shown in Figure 3.

Implementation

Pre-session arrangements

Creation of a large-scale Sudoku grid within a dissection room space created logistical questions. The presence of large mobile whiteboards proved a blank canvas on which to place individual cards. However, this could only be effective if the outlines of the

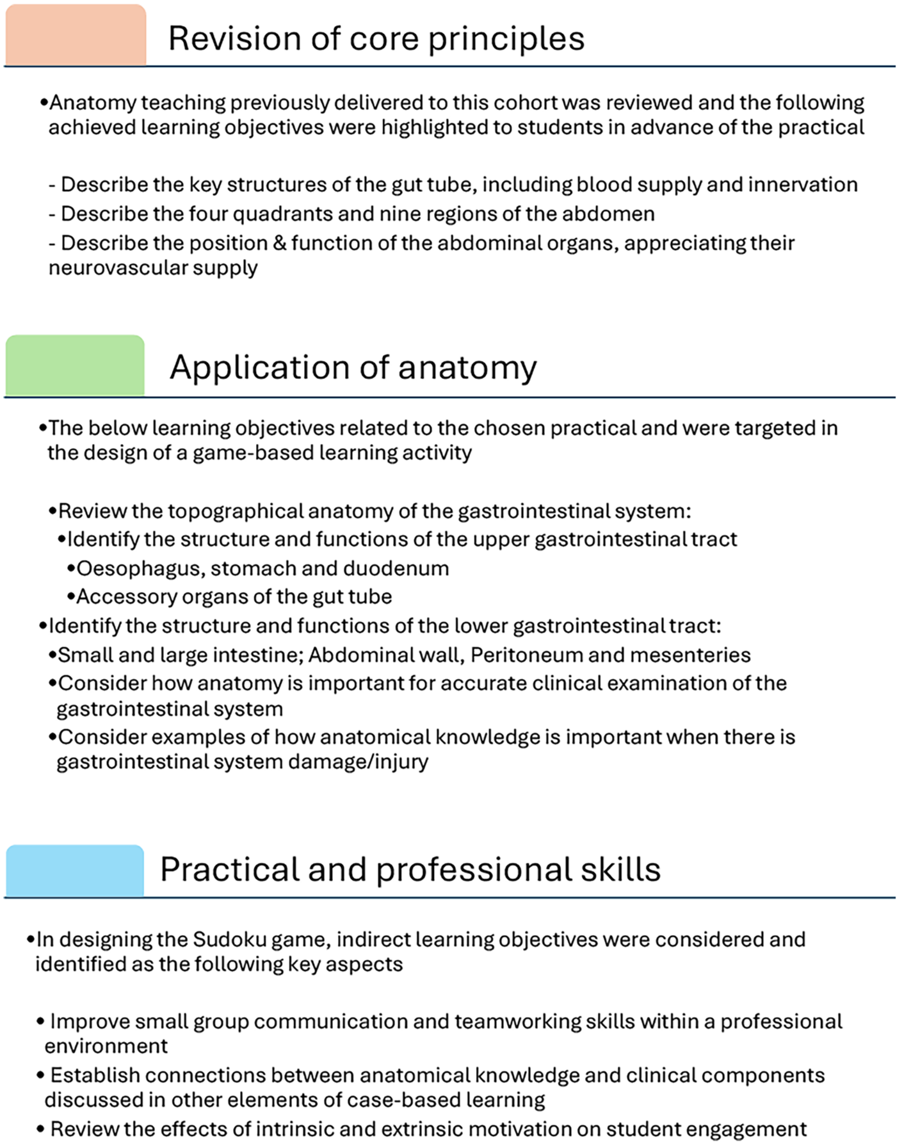

FIGURE 1 Direct and indirect learning objectives of the practical session and game-based learning activity created.

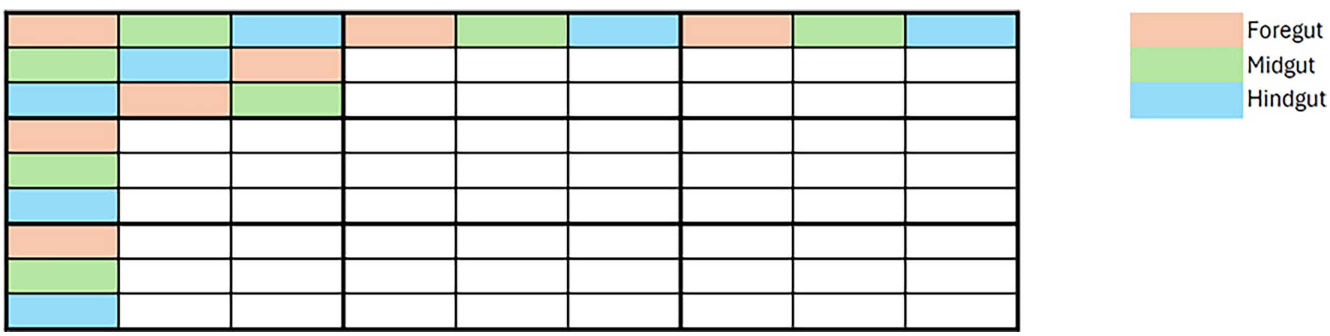

FIGURE 2 Blank Sudoku-style grid containing 81 (9 × 9) cells. The darker lines sort these cells into 9 boxes. The colors are used to depict how each box/row/column might contain three structures related to each embryological subdivision of the gastrointestinal tract.

grid were given in advance and in a format that could not be easily erased, such as drawn on using markers. To achieve this, tape was used to outline the required 81 cells and adhesive provided to attach the prelaminated cards, as shown in Figure 4.

Student instructions

Students were given an introduction to the session on arrival, including a verbal walkthrough of how the game would run. The

Distal	Caecum	Inferior mesenteric ganglion	Pancreas	Superior duodenum	Ileal arteries	Gastroduodenal artery	Appendix	Inferior mesenteric vein
Ileum	Splenic artery	Common hepatic artery	Ileocaecal valve	Suprapubic	Inferior mesenteric lymph nodes	Plicae circularis	Ascending left colic artery	Haematemesis
Coeliac ganglion	Superior mesenteric ganglion	Haematochezia	Transverse colon (3/3)	Middle colic artery	Cystic artery	Vasa recta	Epigastrium	Pelvic splanchnic nerves
Haustra	Left gastric artery	Hepatic flexure	Superior rectal artery	Descending duodenum	Jejunal arteries	Rectosigmoid junction	Major duodenal papilla	Inferior pancreaticoduodenal artery
Stomach	Superior mesenteric artery	Ascending colon	Descending left colic artery	Internal anal sphincter	Vagus	Superior pancreaticoduodenal artery	Sigmoid arteries	Coeliac lymph nodes
Caudal	Splenic flexure	Pyloric sphincter	Spleen	Ileocolic artery	Proximal	Taenia coli	Umbilical	Mesentery
Jejunum	Sigmoid colon	Common bile duct	Transverse colon (2/3)	Right colic artery	Esophagus	Mesentery	Malaena	Vagus
Ascending duodenum	Rectum	Left iliac fossa	Liver	Pancreatic duct	Pectinate line	Haustra	Gallbladder	Superior mesenteric vein
Coeliac artery	Right gastric artery	Inferior duodenum	Transverse colon (1/3)	Inferior mesenteric artery	Anal columns	Hypochondriac	Taenia coli	Descending colon

FIGURE 3 Example completed GrID game. Again, the colors represent the relationship between the feature on the card and to the foregut (orange), midgut (green), or hindgut (blue).

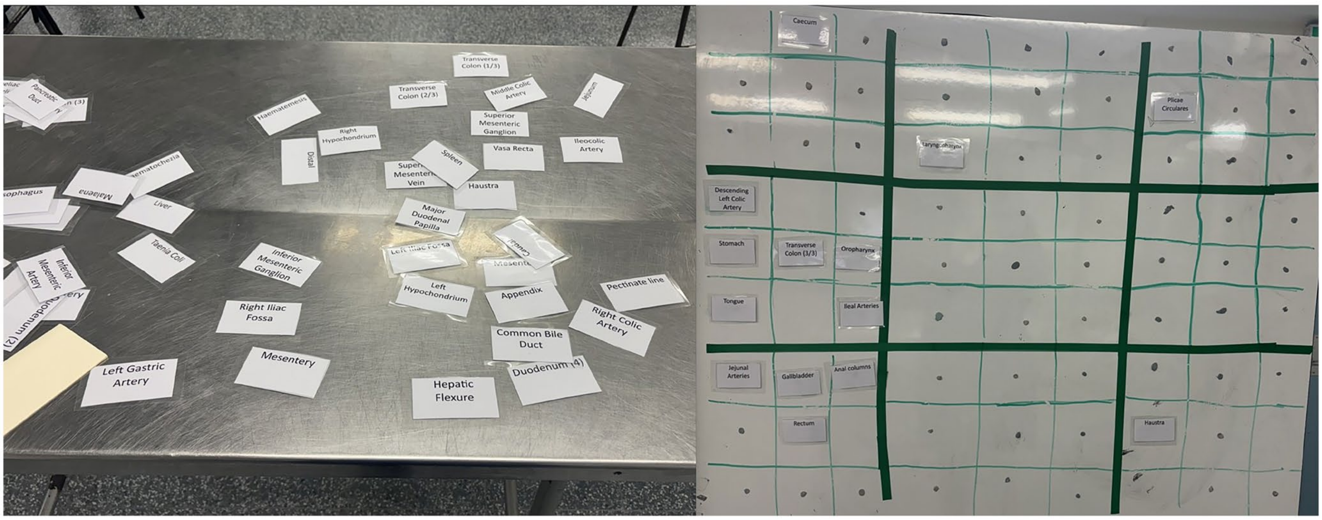

FIGURE 4 Laminated cards were placed on a table for students to read through (left) and attach to the board (right).

opportunity for questions was provided prior to the start of timed rotations. A copy of written instructions was placed with the cards (Appendix A), and the session lead spoke to each group again at the start of their allocated time to ensure the format was understood. Groups were instructed to complete as much of the puzzle as possible during the time given and to look for the hidden code word within. Students were advised to return the sticky notes provided to the session lead with the correct answer and their group number included before they moved on to the next station.

Prize giving

All correct answers were placed in a bowl, which in our dissection room environment was chosen to be a model of a pelvis. To encourage collaborative efforts, each group was allowed to submit only one team answer. At the end of each session, a winning group was randomly drawn and awarded a packet of chocolate digestive biscuits in order to align with the chosen code word and keep the activity low stakes.

REFLECTION AND DISCUSSION

Student engagement

As with most teaching activities, levels of engagement were noted to vary greatly between students. Due to the independent nature of the GrID game, students were not observed throughout this station, but on initial introduction and explanation of the game structure, some students were clearly excited and motivated. Several students gave spontaneous feedback following the session about the positive use of game-based learning within their anatomy teaching and requested similar activities. An asset to this game design is the longevity of Sudoku, as future cohorts are still likely to be familiar with the structure.

Examples of both intrinsic and extrinsic motivation were witnessed during this station. It has been argued that small groups in science classroom spaces are often dominated by students with higher levels of confidence and privilege.[32] While this may still have occurred within the running of this game, it was observed by the session lead that students who expressed an interest in word games and puzzles often took an innate leadership role within this task. These students were intrinsically motivated to do well in this game, as they had a pre-existing connection with the format. This enhanced the learning of both the more motivated students and their peers who benefited from this enthusiasm and engagement. While the subset of students who enjoyed the format could very well have overlapped entirely with those more likely to take a leadership role anyway, this could have also created a new opportunity for students to challenge themselves socially in a protected space with an enjoyable activity.

The potential for a safe learning environment is not a new concept within game design. Salen and Zimmerman introduced the idea of a "magic circle" mindset that posits that game participants exist together in a reality where they can take more risks due to the absence of real-world consequences.[33] The lack of judgment and fear this can afford has been noted as an asset to game use in anatomy,[34] as several anatomical games have noted the beneficial effects of a relaxed environment on student learning.[35] However, as noted by Ang et al.,[6] enjoyable activities may still lead to negative learning outcomes, as links between student satisfaction and knowledge acquisition have not explicitly been drawn in anatomical contexts.

Competition as a motivator

Within anatomical education, there is evidence that use of competition gaming elements can lead to improved student performance and knowledge retention. Van Nuland et al.[36] found a statistical increase in anatomy scores between a group exposed to competitive components early in their studies and those who did not compete. This difference was only observed on the group's final assessment, and earlier comparisons revealed no initial benefit to the first group to use this gaming element. It was posited that the ultimate difference seen could be a result of social comparison during the competition and subsequent individual changes in student study practices. Du et al.[37] also demonstrated that students who competed in a multiplayer version of an educational anatomy game had stronger outcomes compared to single-players. However, variations in motivation may also hinge on the importance anatomy students assign to traditional educational tasks over other activities. Abdel Meguid et al.[38] found that 95% of students placed value on learning the source material, while only 78% preferred activities that spiked curiosity, such as games. The latter group of students is acting out of their own intrinsic motivation to engage, yet extrinsic elements can also be introduced to help bridge this gap and enhance opportunities for competitive learning.

Extrinsic motivation comes from rewards offered—in this case, the packet of digestive biscuits awarded as a prize. It is a short-term motivator and can often paradoxically reduce levels of intrinsic motivation as students readjust their priorities away from learning and focus solely on the successful completion of a task.[5] As a result of this, the use of extrinsic motivators has been advised to be used with caution in game-based learning, and a careful balance needs to be struck between healthy engagement and undue competition and stress.[31]

The third and final cohort to experience the GrID game appeared to be as engaged with the task given as their peers. Despite the witnessed dedicated efforts of team members, there was a noted reduction in correct code word submissions, and some groups were unable to submit an answer at all. At the end of the final practical, individual cards were collected, and it was discovered that several cards that contained bolded letters were missing. Initial reflections from the session lead and demonstrating team present were that these cards may have been accidentally misplaced or unintentionally stored within lab coat pockets. However, these suspicions were quickly altered, as the cards were found scattered purposefully across the dissection room in obscured areas such as behind computers and under equipment. Groups who arrived at the station with the GrID game after these cards were moved were unable to complete the code word and thus did not get included for the prize.

This was an interesting point of reflection for several reasons. The use of prizes within game-based learning has been recommended to remain low-stakes to ensure that competition and stress are minimized.[39] This was considered in the design of the game and as the final practical session took place on a separate day, it was incredibly likely that students present had already heard details of the prizes given to their peers. In which case, these competitive actions were performed irrespective of the degree of prize offered. Another consideration is how these results are viewed alongside the predesignated indirect learning objectives of this activity. Although not expressly stated to students at the onset of the game, teamwork and collaboration were inherent to the format. Student actions that obstructed peer success fall under a definition of sabotage[40] and actively impede opportunities for group effort. Use of competitive actions within this fun activity is out of character with the literature, which suggests that motivations to cheat others are higher within summative landscapes. It may also suggest that these students are more likely to perform academic misconduct in future high-stakes assessments.[41] Furthermore, it calls into question the professionalism of those involved, which is a key tenet of medical education.[42]

However, such actions are never categorical, nor reflective on the activity or cohort as a whole. Feedback from students in similar game-based learning activities has highlighted the enjoyment of competitive elements[43,44] and noted an increase in student motivation.[12,43] Use of similar rewards has also been praised by researchers as a way of enhancing student energy.[6] Reaching an ideal balance between increasing participant motivation and creating undue stress or tactical actions is an ever-present challenge in game-based learning and one that will be continuously reviewed in future use.

Resource adaptability

As a predesigned resource (found in Appendix A), GrID is a low cost and sustainable game to utilize. Once printed and laminated, cards can be retained and used repeatedly with ease. The laminated covering is easy to clean prior to removing from a dissection room with cadaveric material.

The creation of this game was very practical and straightforward once a topic was chosen. Cells were created using tables in Microsoft Word, which was then later replaced with Microsoft Excel for increased ease of use. The rigid nature of Sudoku as a format was beneficial, as all words need to fit within the grid structure and organizational rules. Further expansion into other anatomical systems would help to ascertain the reproducibility of this as a resource, yet components of other systems that could easily be sorted into groups of nine (or even three as used here) are minimal and may require further adaptation. In year groups with more advanced clinical knowledge, common procedures or radiological features could also be incorporated.

Limitations and future use

A key limitation of this reflection is the lack of data and formal student feedback. The initial pilot was included toward the end of delivered anatomy teaching, and it was felt that asking for feedback from students close to summative assessment could result in both low response rates and increased student stress. This practical has yet to be redelivered at time of publication but will aim to be further evaluated to formally assess this game as an educational tool. Further development within alternative human anatomy teaching would also help to robustly review this game, such as within Applied Anatomy undergraduate and Functional and Clinical Anatomy intercalating cohorts.

CONCLUSION

The use of game-based learning within anatomy education is an engaging method to promote collaborative teamwork and the creation of connections between anatomical theory and clinical practice. We have discussed our experiences in designing and implementing a novel Sudoku-based card game targeted at first-year undergraduate students. The resource included is a low-cost, sustainable, and easily reproducible tool for similar use. The novel format creates exciting avenues for further exploration of anatomical concepts through the lens of commonly played number and word games. Subsequent research will focus on a formal review of the efficacy of the GrID game and potential barriers to future adaptation.

AUTHOR CONTRIBUTIONS

Hannah Dunne: Conceptualization; methodology; writing – original draft; writing – review and editing; resources; project administration. **Scott Abbott Paterson:** Writing – review and editing. **Michelle Spear:** Writing – review and editing.

ETHICS STATEMENT

No data has been collected from staff or students involved, and as a result, no ethical approval has been sought for this discursive analysis.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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